

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE – RAIGAD -402 103**

Regular & Supplementary Winter Examination-2023

Branch: Electronics/Electronics and Telecommunication Engineering/Electronics and Communication
Sem.:- V **Course: B. Tech**

Marks: 60

Subject with Subject Code: - BTEXC502/BTETC502 Digital Signal Processing

Date:- 3/1/2024(2 to 5 pm)

Time:3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) State and prove sampling theorem for low pass signals with neat diagram. Also explain the term aliasing.	L2	6
B) A C.T. signal $x(t)$ is obtained at the output of an ideal low pass filter with cut-off frequency $\omega_c = 1000\pi$ rad/sec. If instantaneous sampling is performed on $x(t)$, which of the following sampling period would guarantee that $x(t)$ can be recovered from its sampled version using an appropriate low pass filter. 1) $T_s = 0.5 \times 10^{-3}$ 2) $T_s = 2 \times 10^{-3}$ 3) $T_s = 10^{-4}$	L3	6
C) Write the applications of Digital Signal Processing.	L1	6
Q.2 Solve Any Two of the following.		12
A) State and prove the following properties of DTFT: 1) Frequency shifting 2) Time shifting 3) Differentiation in frequency 4) Time convolution	L1	6
B) Determine and sketch the magnitude and phase response of $y(n)$: $y(n) = 0.5\{x(n) + x(n-2)\}$	L2	6
C) Determine N-point IDFT of $X(k) = \delta(k)$, $0 \leq k \leq N-1$ and for $0 \leq k_0 \leq N-1$, determine the DFT of using the previous result. 1) $x_1(n) = e^{j\frac{2\pi}{N}k_0n}$ 2) $x_2(n) = e^{-j\frac{2\pi}{N}k_0n}$ 3) $x_3(n) = \cos\left(\frac{2\pi}{N}k_0n\right)$ 4) $x_4(n) = \sin\left(\frac{2\pi}{N}k_0n\right)$	L2	6

Q.3 Solve Any Two of the following. **12**

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| A) Find the linear convolution of the following two signals using overlap and save method.
$x(n) = \{3, -1, 0, 1, 3, 2, 0, 3, 2, 2\}$ and $h(n) = \{1, 1, 1\}$ | L2 | 6 |
| B) Complete the convolution of the convolution property of Z-Transform.
$x_1(n) = \{1, -2, 1\}$, $0 \leq n \leq 2$
and $x_2(n) = \{1, 1, 1, 1, 1, 1\}$, $0 \leq n \leq 5$ | L2 | 6 |
| C) Determine the Z.T. and ROC of the following finite duration signals:
(a) $x[n] = \{1, 2, 3, -1, 0, 1\}$, $-2 \leq n \leq 3$
(b) $x[n] = \{0, 0, 1, 2, 1\}$, $0 \leq n \leq 4$
(c) $x[n] = \{1, 2, 3, -1, 0\}$, $-4 \leq n \leq 0$ | L1 | 6 |

Q.4 Solve Any Two of the following. **12**

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| A) Design a simple Low Pass FIR digital filter. Plot its pole zero plot. Obtain and plot magnitude and phase response for the same. | L3 | 6 |
| B) Derive the expression to obtain the order of a Butterworth filter. | L1 | 6 |
| C) Derive expression of $y(n)$ and draw the direct form and its transposed structure of the transfer function given by $H(z) = 1 - (1/3)z^{-1} + (1/6)z^{-2} + z^{-3}$. | L2 | 6 |

Q.5 Solve Any Two of the following. **12**

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|---|----|---|
| A) Explain the indirect method of designing digital filters from analog filters. Also differentiate between digital filters and analog filters. | L1 | 6 |
| B) Explain Aliasing effect in downsampling with the help of neat diagrams. Explain to the process to prevent this effect. | L2 | 6 |
| C) For the transfer function | L3 | 6 |

$$H(s) = \frac{1}{s^2 + \sqrt{2}s + 1}$$

Find the corresponding system function $H(z)$ using bilinear transformation algorithm. Assume $T=1$.

*** End ***